CS & IT ENGINEERING



Database Management System

Relational model and Normal forms



DPP-03

Discussion Notes

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[MCQ]

#Q.Assume a relation schema R with 5 attributes P, Q, R, S, T and the set of FD'S $P \rightarrow RS$, $Q \rightarrow RT$, $T \rightarrow Q$ consider the statements:

S_1: The only candidate keys of R are PQ and PT

S_2: The highest normal form satisfied by R is 2NF

Which of the statement is true?

$PQ^+ = \{P, Q, R, S, T\}$

PT^+

P

A Only S_1 is true

B Only S_2 is true

C Both S_1 and S_2 are true

D Neither S_1 and S_2 is true

[MCQ]

#Q.Assume a relation R = (P, Q, R, S) and a set F of functional dependencies:

F = {PR → S, S → P, S → Q, S → R}, Highest normal form satisfied by the relation R is?

S⁺ = { P, Q, R, S }

PR⁺

A 2NF

B 3NF

C BCNF

D 1NF

[MCQ]

#Q.Consider a relation R(P, Q, R, S, T, U, V, W) with the following functional dependencies:

{RW→V, P → QR, Q → RUW, T → P, U → TV}, then highest normal form of the relation R is _____.

A 1NF

B 2NF

C 3NF

D BCNF

$P\ S^+ = \{P, Q, R, S, U, W, T, V\}$

TS, US, QS

[MCQ]

#Q.Consider a table/Relation R has only one candidate key, then which of the following is always true?

A. If R is in 2NF, then it is also in 3NF.

B. If R is in 3NF, then it is also in BCNF.

C. If R is in 2NF, but it is not in 3NF.

D. None of the above.

[MCQ]

#Q.Consider a relation R(P, Q, R, S, T) with the set of FD's

{PQR →ST and T → QRS} which of the following statements is true?

A. R is not in 2NF

B. R is in 2NF but not in 3NF

C. R is in 3NF but not in BCNF

D. R is in BCNF

$(PQR)^+ = \{P, Q, R, S, T\}$

↓

PT

↓

T → S

[MCQ]

#Q.Consider a relation R (L, M, N, O) with the functional dependencies:

$L \rightarrow M$,
$M \rightarrow N$,
$N \rightarrow O$

Which one of the following decompositions is not lossless?

A R_1 (L, M), R_2(M, N), R_3 (N, O)

B R_1(L, M), R_2(L, N), R_3 (L, O)

C R_1(L, O), R_2(M, O), R_3 (N, O)

D All of the above are lossless

[MCQ]

#Q.Suppose functional dependency Q → R holds in relation R (P, Q, R, S) which additional FD will make R be in 3NF, but not BCNF?

A. S → PQ

B. PR → S

C. RS → Q

D. PS → Q

$A = \{ Q \to R, \; S \to PQ \}$ (Q → R: 3NF; S → PQ: BCNF)

$S^+ = \{ S, P, Q, R \}$

$B = \{ Q \to R, \; PR \to S \}$ (Q → R: PSCK; PR → S: NPA)

$PQ^+ = \{ P, Q, R, S \}$

$C = \{ Q \to R, \; RS \to Q \}$ (3NF; Q → R: PSCK; RS → Q: PA)

$PS^+ = \{ P, S \}$

$PQS^+ = \{ P, Q, R, S \}$

PRS

$D = \{ Q \to R, \; PS \to Q \}$ (2NF; PS → Q: BCNF)

$PS^+ = \{ P, S, Q, R \}$

[MCQ]

#Q.Consider a relation schema S = { A, B, C, D, E, F, G, H } with the following set of functional dependencies:

$AB \rightarrow E$

$A \rightarrow CF$

$B \rightarrow D$

$D \rightarrow GH$

$B^+ = \{ B, D, G, H, \}$

#Q.Now, consider the following decompositions of S:

Decomposition 1: $D_1 = \{ S_1, S_2, S_3, S_4 \}$

$S_1 = \{ A, B, C \}$

$S_2 = \{ A, D, F \}$

$S_3 = \{ B, E \}$

$S_4 = \{ D, G, H \}$

Decomposition 2: $D_2 = \{ S_1, S_2, S_3, S_4 \}$

$S_1 = \{ A, B, C \}$

$S_2 = \{ D, F \}$

$S_3 = \{ B, E \}$

$S_4 = \{ D, G, H \}$

Which of the above decompositions has the lossless join property?

A. Only D_1

B. Only D_2

C. Both D_1 and D_2

D. Neither D_1 nor D_2

[MCQ]

#Q.Which of the following statements is FALSE ?

R(A, B)

AB is CK

$P \rightarrow \overline{Q} \equiv T$

P → F

A Any relation with 2 attributes is in BCNF. True

B A relation in which every key has only one attribute is in 2NF → TRUE → PSCK → NPA (X)

C A prime attribute can be transitively dependent on a key in a 3NF relation. TRUE

D A prime attribute can be transitively dependent on a key in a BCNF relation.

[MCQ]

#Q.A functional dependency of the form P → Q is trivial if:

↓

P ⊇ Q

A $Q \subseteq P$

B $P \subset Q$

C $P \subseteq Q$

D $P \subseteq Q$ and $Q \subseteq P$

THANK - YOU

